1. MLP: parameter calculation

Lab 3 - Task 3 | G M Robi Farhan Rojob | Student number: 73997

Counting convention: count every trainable weight and bias. Output values are activations, not additional learned parameters. All three models produce **10 output scores per image**, one for each Fashion-MNIST class.

Architecture: Flatten 28 x 28 = 784 inputs; Linear 784 to 256; ReLU; Linear 256 to 128; ReLU; Linear 128 to 10.

Formula for a linear layer

Parameters = input features x output features + output features

The first term counts weights. The second counts one bias for each output.

Layer 1: Linear 784 to 256

Weights = 784 x 256 = 200,704

Biases = 256

P_1 = 200,704 + 256 = **200,960**

Output: 256 values per image.

Layer 2: Linear 256 to 128

Weights = 256 x 128 = 32,768

Biases = 128

P_2 = 32,768 + 128 = **32,896**

Output: 128 values per image.

Output layer: Linear 128 to 10

Weights = 128 x 10 = 1,280

Biases = 10

P_3 = 1,280 + 10 = **1,290**

Output: 10 values per image. Flatten and ReLU have 0 trainable parameters.

> **Total MLP parameters**
> P = 200,960 + 32,896 + 1,290 = **235,146**

Reference: supplied textbook, section 4.2; Listing 4.2, printed p. 156 (PDF viewer page 208).

2. CNN: output sizes and parameters

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Architecture: Conv(1 to 16, 3 x 3, padding 1); ReLU; MaxPool 2; Conv(16 to 32, 3 x 3, padding 1); ReLU; MaxPool 2; Flatten; Linear 1568 to 10.

Formulas

$$\text{Conv parameters} = (K \times K \times C_{in} + 1) \times C_{out}$$

$$H_{out} = \text{floor}((H_{in} + 2p - K) / s) + 1$$

Here K = 3, padding p = 1, stride s = 1, dilation = 1. The same output-size formula applies to width.

Convolution 1: input 1 x 28 x 28

$$H_{out} = \text{floor}((28 + 2 - 3) / 1) + 1 = 28$$

Weights = 3 x 3 x 1 x 16 = 144; biases = 16

$$P_{conv1} = 144 + 16 = \mathbf{160}$$

Conv output = **16 x 28 x 28** = 12,544 values.
After MaxPool 2: **16 x 14 x 14** = 3,136 values.

Convolution 2: input 16 x 14 x 14

$$H_{out} = \text{floor}((14 + 2 - 3) / 1) + 1 = 14$$

Weights = 3 x 3 x 16 x 32 = 4,608; biases = 32

$$P_{conv2} = 4{,}608 + 32 = \mathbf{4{,}640}$$

Conv output = **32 x 14 x 14** = 6,272 values.
After MaxPool 2: **32 x 7 x 7** = 1,568 values.

Output layer: Flatten, then Linear 1568 to 10

Input features = 32 x 7 x 7 = 1,568

Weights = 1,568 x 10 = 15,680; biases = 10

$$P_{output} = 15{,}680 + 10 = \mathbf{15{,}690}$$

Final output = **10 values per image**. ReLU, MaxPool and Flatten have 0 trainable parameters.

> **Total CNN parameters**
> P = 160 + 4,640 + 15,690 = **20,490**

Why not multiply parameters by image height and width? The same filter weights are shared across every spatial position.

Reference: supplied textbook, section 4.3; Listing 4.4, printed p. 161 (PDF viewer page 213).

3. Attention: parameter calculation

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Image = 28 x 28; patch = 7 x 7; number of patches = (28/7) x (28/7) = **16**. Each patch has **49** values. Embedding d = 64; heads h = 4; dimension per head = 64/4 = 16; feedforward width = 128; encoder layers = 2.

A. Patch projection: Linear 49 to 64

P_{patch} = 49 x 64 + 64 = 3,136 + 64 = **3,200**

Output = **16 x 64** per image. This projection is shared by all 16 patches; its parameter count is not multiplied by 16.

B. Parameters inside ONE Transformer encoder layer

Query: P_Q = 64 x 64 + 64 = **4,160**

Key: P_K = 64 x 64 + 64 = **4,160**

Value: P_V = 64 x 64 + 64 = **4,160**

QKV total = 3 x 4,160 = **12,480**

Attention output projection = 64 x 64 + 64 = **4,160**

Feedforward 1: 64 x 128 + 128 = **8,320**

Feedforward 2: 128 x 64 + 64 = **8,256**

Each LayerNorm has 64 learned scale values and 64 learned shift values. There are two LayerNorms in each encoder.

LayerNorm total = 2 x (64 + 64) = **256**

$P_{encoder}$ = 12,480 + 4,160 + 8,320 + 8,256 + 256

$P_{encoder}$ = **33,472**

Four heads split the 64 features. They do not create four separate 64-feature QKV projections, so **do not multiply 12,480 by 4**.

C. Two encoder layers, mean pooling and classifier

Two encoders = 2 x 33,472 = **66,944**

Output remains **16 x 64**. Mean pooling gives **64** values and has 0 trainable parameters.

Final Linear 64 to 10: $P_{classifier}$ = 64 x 10 + 10 = **650**

Total attention-model parameters
P = 3,200 + 66,944 + 650 = **70,794**

Final output = **10 scores**. ReLU, dropout, attention softmax and residual addition add no trainable parameters. No positional embedding or class token is specified or included.

Reference: supplied textbook, section 4.5, printed pp. 169-176 (PDF pages 221-228); counts include the two affine LayerNorms in each implemented PyTorch encoder.

4. Final totals and Task 3 FLOPs

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Parameter and output summary

Model	All trainable parameters	Final classifier parameters	Final outputs per image
MLP	235,146	1,290	10
CNN	20,490	15,690	10
Attention	70,794	650	10

Parameters and outputs are different quantities. A batch of 128 produces 128 x 10 = 1,280 scores. The model's learned parameter count stays unchanged.

The exact layer specifications give the totals above. They conflict with the manual's 100k-150k target; the models have not been resized.

Additional handwritten requirement: representative FLOPs

The manual also asks for FLOPs. For one image, count one multiplication plus one addition as 2 FLOPs. Below, count matrix/conv arithmetic only; omit bias additions, activation, pooling, normalization, softmax and residual operations.

MLP

First linear: 2 x 784 x 256 = **401,408 FLOPs**

Whole MLP: 401,408 + 65,536 + 2,560 = **469,504**

CNN

First conv: 2 x 3 x 3 x 1 x 16 x 28 x 28 = **225,792**

Second conv: 2 x 3 x 3 x 16 x 32 x 14 x 14 = **1,806,336**

Final linear: 2 x 1,568 x 10 = **31,360**

Whole CNN: 225,792 + 1,806,336 + 31,360 = **2,063,488**

Attention: sequence L = 16, embedding d = 64

QKV: 2 x 16 x 64 x 3 x 64 = **393,216**

Scores QK^T: 2 x 16 x 16 x 64 = **32,768**

Attention times V: 2 x 16 x 16 x 64 = **32,768**

Output projection: 2 x 16 x 64 x 64 = **131,072**

Two feedforward products: 2 x (2 x 16 x 64 x 128) = **524,288**

One encoder = 393,216 + 32,768 + 32,768 + 131,072 + 524,288

One encoder = **1,114,112 FLOPs**

Whole attention = (2 x 16 x 49 x 64) + 2 x 1,114,112 + (2 x 64 x 10)

Whole attention = 100,352 + 2,228,224 + 1,280 = **2,329,856**

Source: Lab Manual, Lab 3, printed p. 5. These are the exact models used in the completed lab; parameter totals agree with torchinfo.summary. The full profiler comparison and timing evidence remain in report.md.